

Image Processing Pipeline

MNIST PNG Dataset - CS.383 Group Project

1. Project Overview

This project implements a four-stage image processing pipeline on the MNIST handwritten digit dataset (PNG format). The dataset is loaded from a zip file uploaded to Google Colab. Each stage builds on the previous one: dataset collection, preprocessing, filtering, and edge detection with segmentation. Each stage is assigned to one team member. The project is part of CS.383.

2. Project Structure

File	Description
Project Image.ipynb	Main Colab notebook - runs all 4 stages end-to-end
documentation.pdf	Project report and documentation
README.md	Setup and usage instructions

3. Pipeline Stages

Person 1 - Dataset Collection

- Extracts dataset from uploaded zip file (/content/test (2).zip)
- Scans digit subfolders 0-9 and reports image counts per class
- Displays one sample per digit in a 2x5 matplotlib grid

Person 2 - Preprocessing

- Resizes images to 28x28 pixels using anti-aliasing
- Converts RGB to grayscale via `skimage.color.rgb2gray`
- Normalizes pixel values to the range [0, 1]
- Applies intensity inversion: $\text{image_inv} = 1.0 - \text{image_norm}$

Person 3 - Image Filtering

- Gaussian Blur - smooths noise, Gaussian kernel ($\sigma=1$)
- Median Filter - removes salt-and-pepper noise, disk element ($\text{radius}=1$)
- Sharpening - 3x3 Laplacian-based kernel via `cv2.filter2D`

Person 4 - Edge Detection & Segmentation

- Sobel - gradient magnitude in X and Y directions
- Canny - multi-stage edge detector (thresholds: 50 / 150)
- Otsu Threshold - automatic binary segmentation (`cv2.THRESH_OTSU`)

4. Dataset

The dataset is loaded from a zip file uploaded directly to Google Colab. No external download or API key is required. The zip file must be uploaded to /content/ before running the notebook. It extracts to a folder named 'test/' containing one subfolder per digit class.

Default zip path (edit in the first cell if needed):

```
zip_path = "/content/test (2).zip"
```

Dataset image counts (MNIST test set):

Digit	Image Count
0	980
1	1135
2	1032
3	1010
4	982
5	892
6	958
7	1028
8	974
9	1009

Total: 10,000 images

5. Requirements

All libraries are pre-installed on Google Colab. If needed, run in a cell:

```
!pip install scikit-image opencv-python-headless matplotlib numpy
```

Library	Used For
scikit-image	Image I/O, resize, grayscale conversion, Gaussian/median filtering
opencv-python	Sharpening kernel, Sobel, Canny, Otsu thresholding
matplotlib	Inline figure output at each stage
numpy	Array operations and type conversions

6. Usage - Google Colab

Step 1 - Open the notebook

Go to colab.research.google.com and upload Project Image.ipynb via File > Upload notebook.

Step 2 - Upload the dataset zip

Click the folder icon in the left sidebar and upload your zip file. The notebook expects:

```
zip_path = "/content/test (2).zip"
```

Step 3 - Run all cells

Use Runtime > Run all, or run each cell in order.

7. Expected Output

Stage	Owner	Matplotlib Figure
Dataset Collection	Person 1	2x5 grid - one sample digit image per class (0-9)
Preprocessing	Person 2	4 panels: Original / Resized / Normalized / Inverted
Filtering	Person 3	4 panels: Grayscale / Gaussian / Median / Sharpened
Edge Detection	Person 4	4 panels: Grayscale / Sobel / Canny / Otsu

The notebook ends with the message: checkmark Full pipeline complete!

8. How to Test the Project

Follow these steps to verify the notebook runs correctly:

- Step 1 Upload Project Image.ipynb to Google Colab
- Step 2 Upload the dataset zip to /content/ in the Colab file browser
- Step 3 Run all cells (Runtime > Run all)
- Step 4 Cell output shows digit counts for classes 0-9
- Step 5 Person 1 figure: 2x5 grid of sample digit images
- Step 6 Person 2 figure: 4 panels showing preprocessing steps
- Step 7 Person 3 figure: 4 panels showing filtering results
- Step 8 Person 4 figure: 4 panels showing edge detection results
- Step 9 Final output: checkmark Full pipeline complete!